

Experiment Report

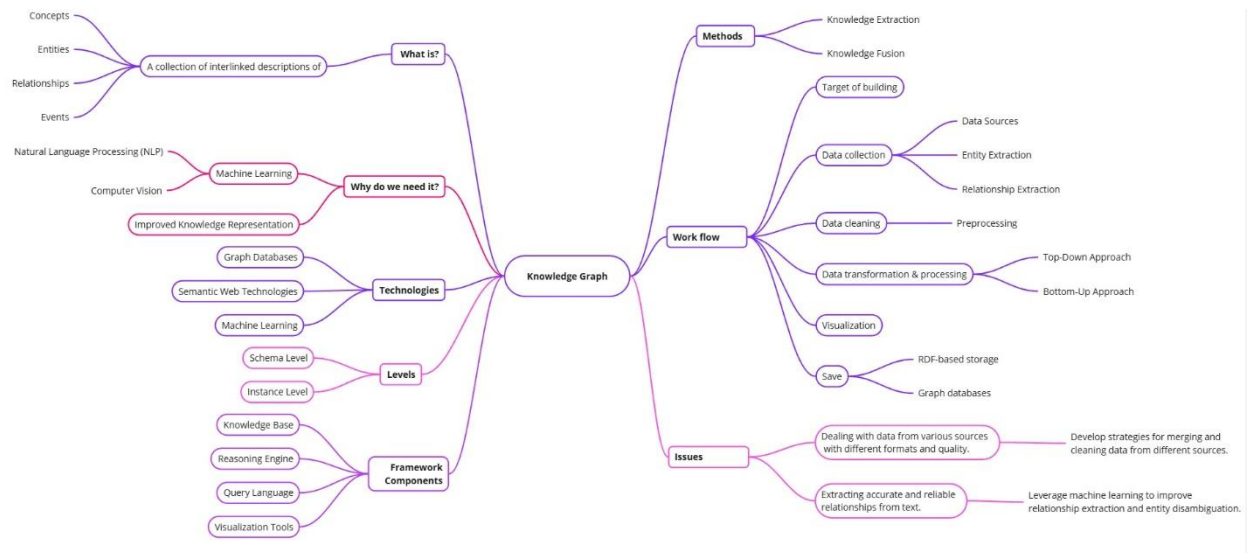
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Course: Big Data Analysis Technology

Experiment Topic: Knowledge Graph

Mind Map Knowledge Graph



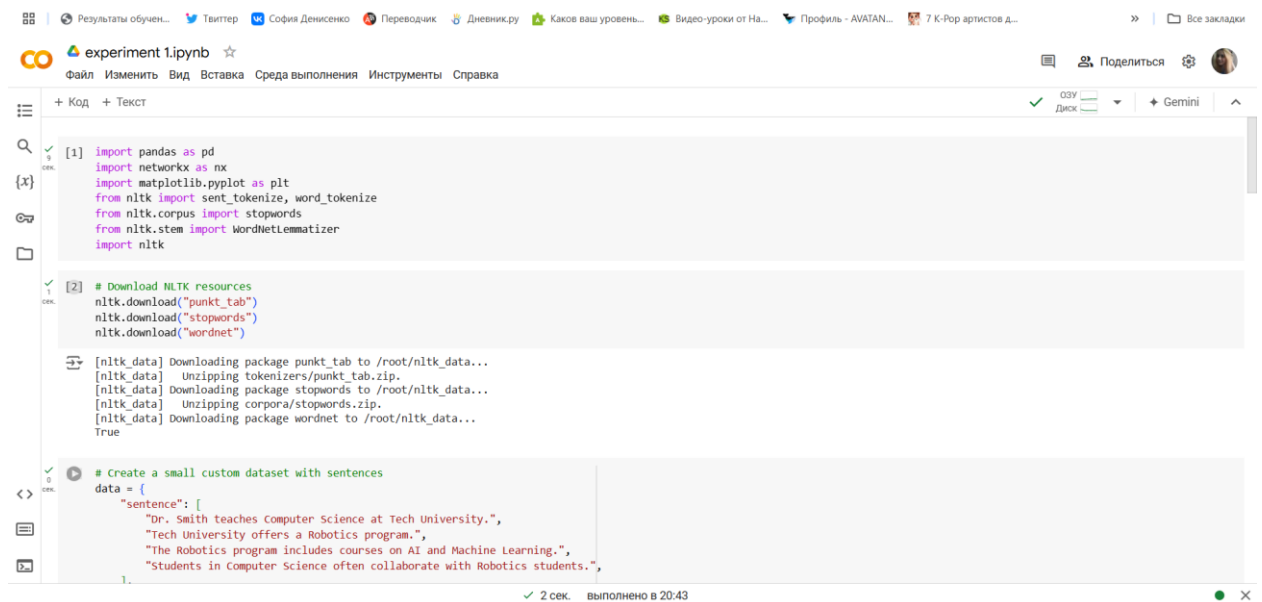
Experiment 1

Experiment Topic: Building a Knowledge Graph in NLP.

Task: Complete the experiment by adding at least 1 new edge and node to the graph beyond the example provided.

I decided to do the experiment in Google Colab.

Opening file with the code and running it.



```
[1] import pandas as pd
import networkx as nx
import matplotlib.pyplot as plt
from nltk import sent_tokenize, word_tokenize
from nltk.corpus import stopwords
from nltk.stem import WordNetLemmatizer
import nltk

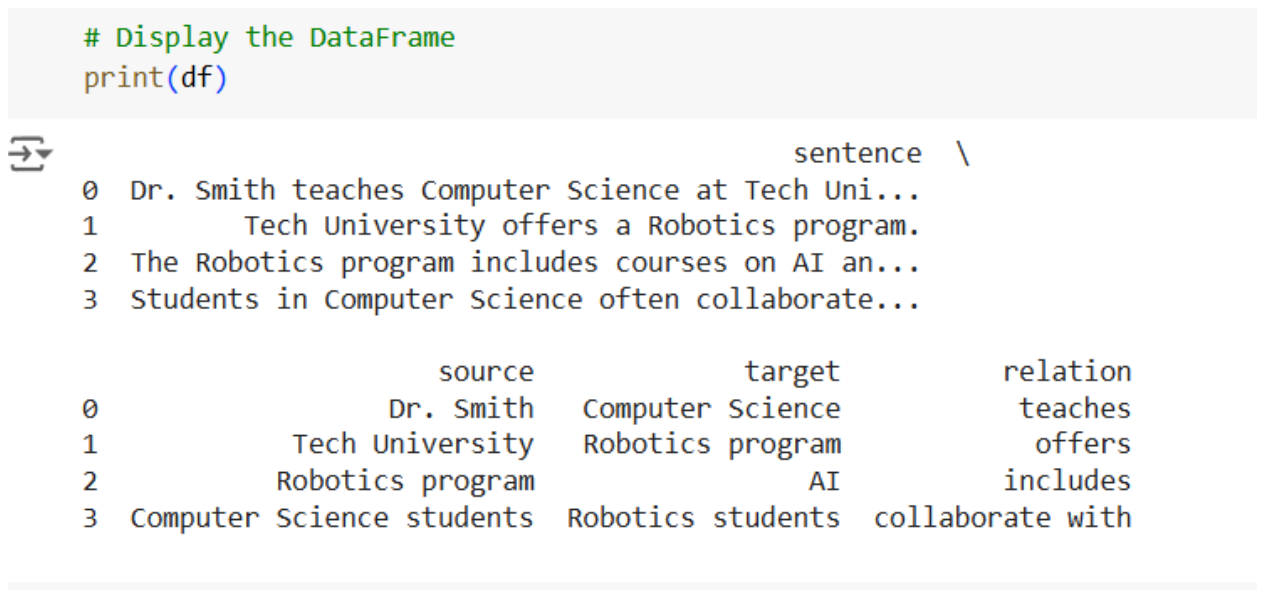
[2] # Download NLTK resources
nltk.download("punkt_tab")
nltk.download("stopwords")
nltk.download("wordnet")

[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data] Unzipping tokenizers/punkt_tab.zip.
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data] Unzipping corpora/stopwords.zip.
[nltk_data] Downloading package wordnet to /root/nltk_data...
True

[3] # Create a small custom dataset with sentences
data = {
    "sentence": [
        "Dr. Smith teaches Computer Science at Tech University.",
        "Tech University offers a Robotics program.",
        "The Robotics program includes courses on AI and Machine Learning.",
        "Students in Computer Science often collaborate with Robotics students."
    ]
}
```

Results:

Dataset:



```
# Display the DataFrame
print(df)
```

	sentence \	source	target	relation
0	Dr. Smith teaches Computer Science at Tech Uni...	Dr. Smith	Computer Science	teaches
1	Tech University offers a Robotics program.	Tech University	Robotics program	offers
2	The Robotics program includes courses on AI an...	Robotics program	AI	includes
3	Students in Computer Science often collaborate...	Computer Science students	Robotics students	collaborate with

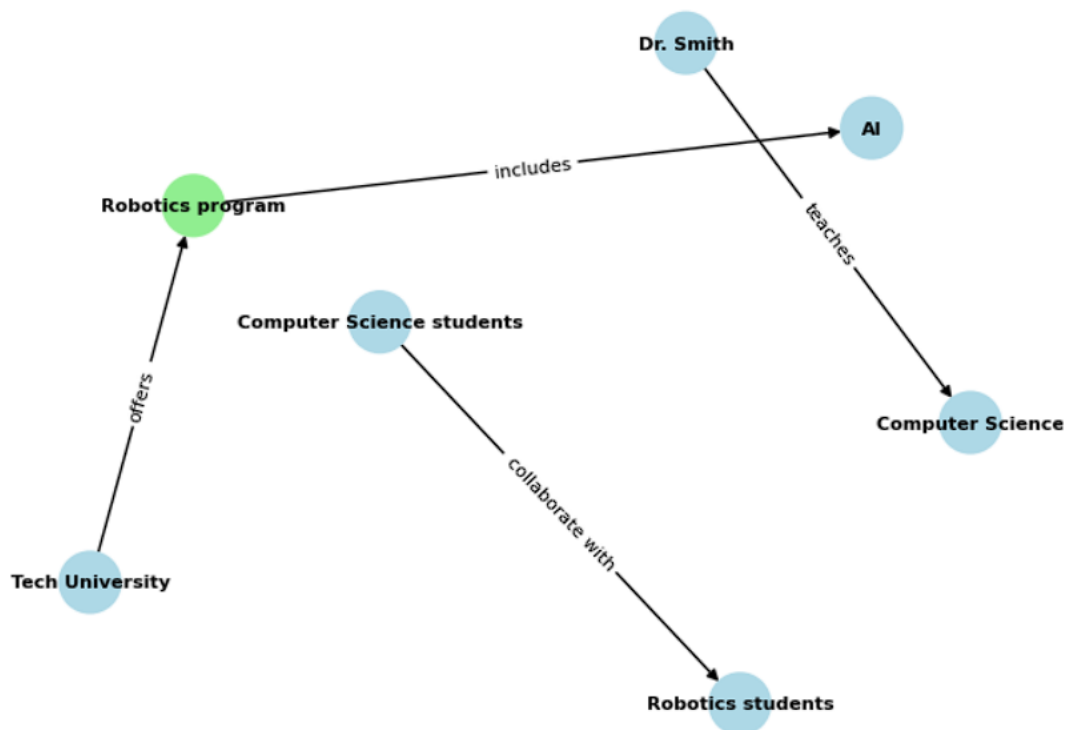
Preprocessing sentences:

```
[link_data] = unzipping_tokenizers/punkc.zip.
                                sentence \
0 Dr. Smith teaches Computer Science at Tech Uni...
1 Tech University offers a Robotics program.
2 The Robotics program includes courses on AI an...
3 Students in Computer Science often collaborate...

                                source          target          relation \
0 Dr. Smith          Computer Science          teaches
1 Tech University    Robotics program          offers
2 Robotics program    AI                      includes
3 Computer Science students  Robotics students collaborate with

                                processed_sentence
0 smith teach computer science tech university
1 tech university offer robotics program
2 robotics program includes course ai machine le...
3 student computer science often collaborate rob...
```

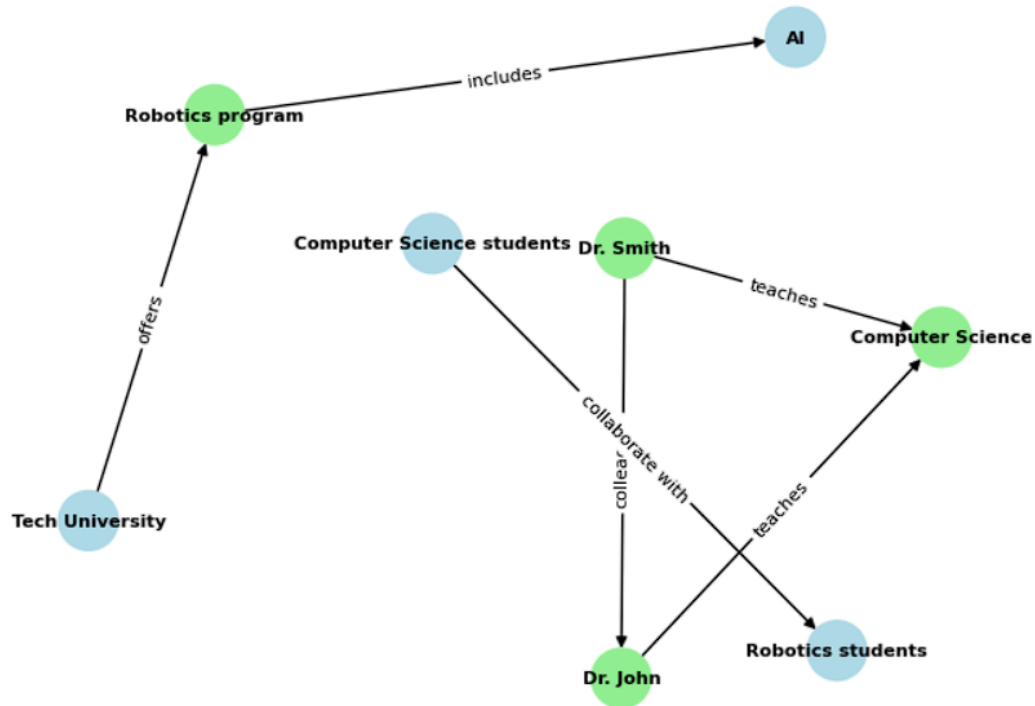
Resulting graph:



Adding more edges and nodes:

```
[14] G.add_edges_from([
    ('Dr. John', 'Computer Science', {'relation': 'teaches'}),
    ('Dr. Smith', 'Dr. John', {'relation': 'colleague'})
])
```

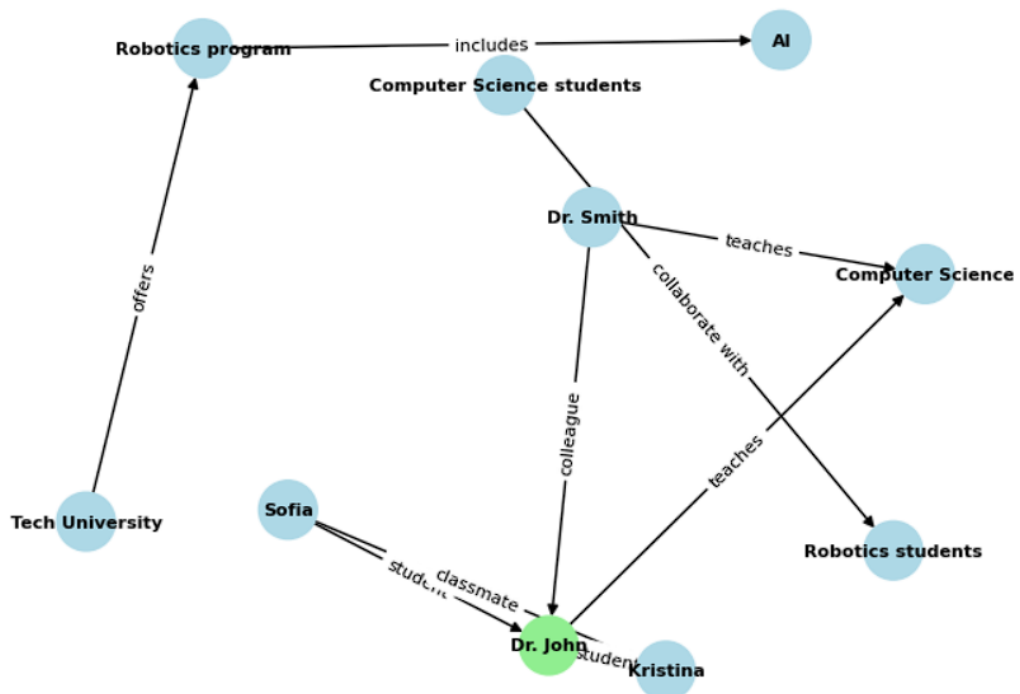
Resulting Graph:



For the task I decided to add two students of Dr. John. And

```
[16] G.add_edges_from([
    ('Sofia', 'Dr. John', {'relation': 'student'}),
    ('Kristina', 'Dr. John', {'relation': 'student'}),
    ('Sofia', 'Kristina', {'relation': 'classmate'})
])
```

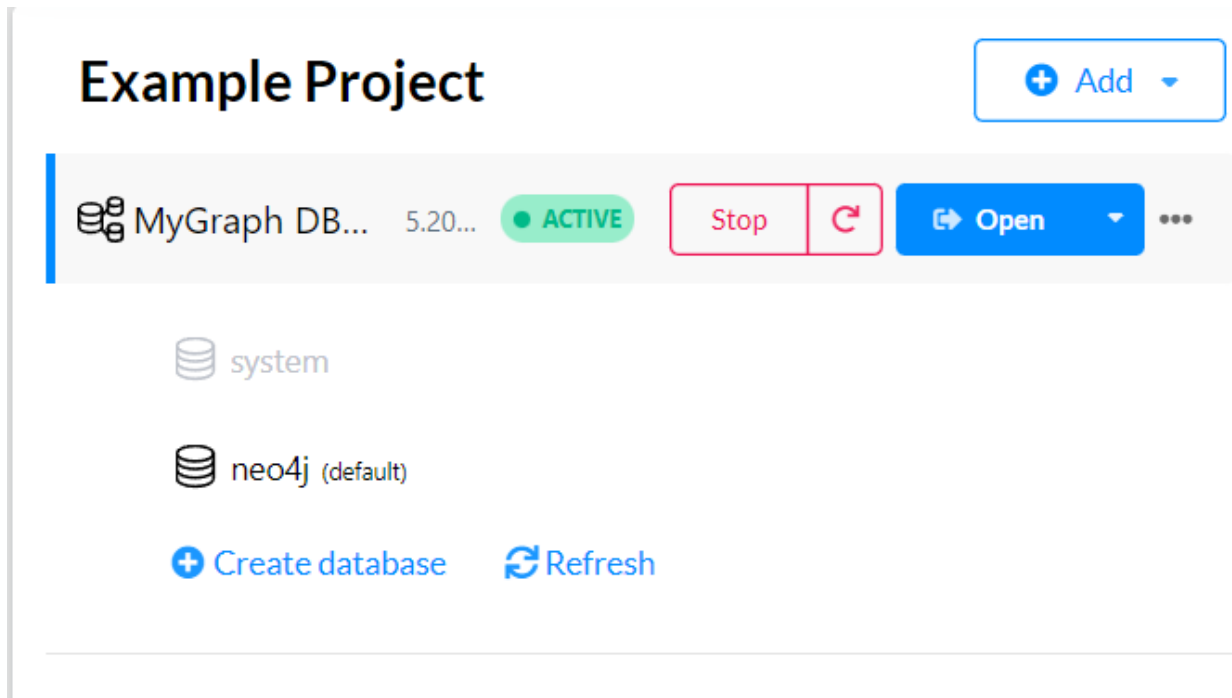
Resulting graph:



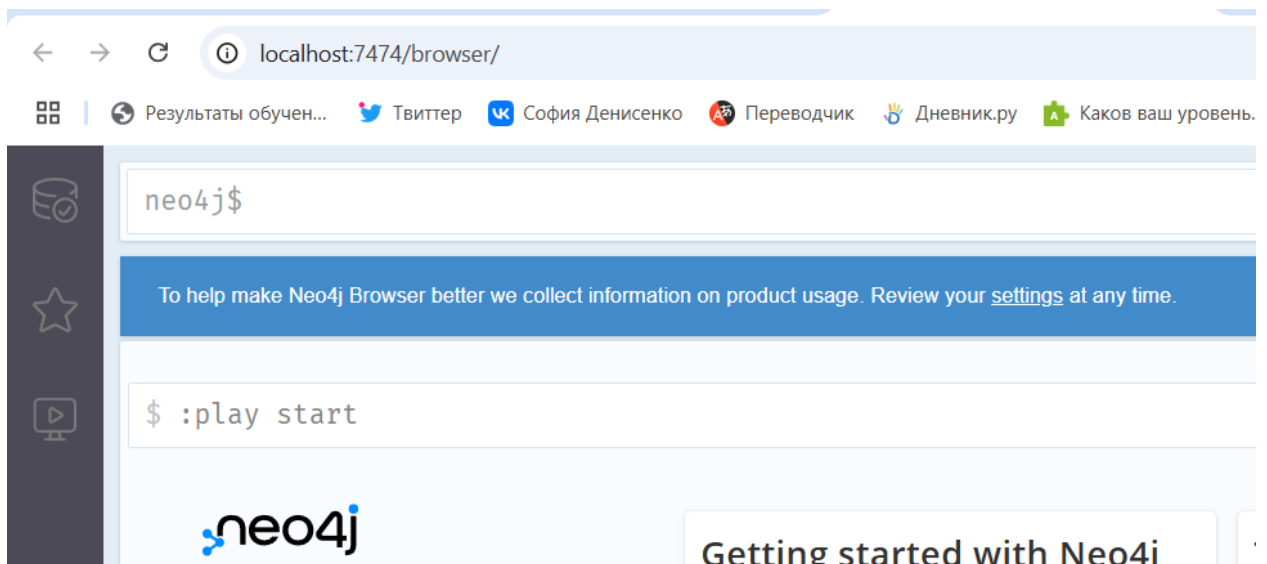
Experiment 2

Experiment Topic: Creating a Movie Graph Database in Neo4j.

Starting Neo4j.



Opening the neo4j browser by connecting to <http://localhost:7474/browser/>.



Pasting the code in PyCharm and running it.

```
1 from neo4j import GraphDatabase
2
3 # Connection details
4 uri = "bolt://localhost:7687" # Default URI for Neo4j
5 username = "neo4j" # Default username
6 password = "00000000" # My Neo4j password
7
8 # Create a driver instance
9 driver = GraphDatabase.driver(uri, auth=(username, password))
10
11 # Usage
12 def create_graph(tx):
13     # Create Movie nodes
14     tx.run("CREATE (:Movie {name: 'Inception'})")
15     tx.run("CREATE (:Movie {name: 'The Dark Knight'})")
16     tx.run("CREATE (:Movie {name: 'Interstellar'})")
17     tx.run("CREATE (:Movie {name: 'Pulp Fiction'})")
18
19 create_graph()
```

Run console output:

```
<Record m.name='The Matrix' type(r)='BELONGS_TO' g.name='Sci-Fi'>
<Record m.name='Inception' type(r)='BELONGS_TO' g.name='Sci-Fi'>
<Record m.name='The Dark Knight' type(r)='BELONGS_TO' g.name='Action'>
<Record m.name='Interstellar' type(r)='BELONGS_TO' g.name='Drama'>
<Record m.name='Pulp Fiction' type(r)='BELONGS_TO' g.name='Thriller'>
```

Results:

```
C:\Users\prett\PycharmProjects\rucharm\venv\Scripts\python.exe "C:\Users\prett\PycharmProjects\rucharm\experiment 2.py"
<Record m.name='The Matrix' type(r)='BELONGS_TO' g.name='Sci-Fi'>
<Record m.name='Inception' type(r)='BELONGS_TO' g.name='Sci-Fi'>
<Record m.name='The Dark Knight' type(r)='BELONGS_TO' g.name='Action'>
<Record m.name='Interstellar' type(r)='BELONGS_TO' g.name='Drama'>
<Record m.name='Pulp Fiction' type(r)='BELONGS_TO' g.name='Thriller'>
Process finished with exit code 0
```

In Neo4j the resulting Movie Graph Database looks like this:

